# Structural Transformation Towards Healthcare

**by**

**Fil Babalievsky**
**U.S. Census Bureau**

**CES 26-62 September 2026**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

Papers in the series are posted on www.census.gov/library/working-papers/series/ces-wp.html. For information about the series, contact Danielle H. Sandler, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov.

**Abstract**

This paper studies variation in healthcare expenditures across time and space using a model of structural transformation tailored to the healthcare sector. It addresses a key tension: cross-country and time-series data show the health spending share rising with overall consumption, suggesting an aggregate income elasticity above one, yet within countries health spending is flat across household incomes, and quasi-experimental spending responses to oil-driven income shocks rise less than one-for-one with income. The calibrated model reconciles this tension in two ways: First, health spending is equalized across incomes by the government or insurers. Second, with an own-price elasticity greater than one in absolute value, positive shocks outside healthcare such as oil income shocks generate price effects that drive the health share down. I find that rising incomes are indeed the dominant driver of healthcare spending, and that reforms that raise the productivity of the healthcare sector will drive its spending share up.

**Keyword:** Healthcare, Structural Transformation

**JEL Classification:** I15, O11

* US Census Bureau, 4600 Silver Hill Road, Suitland, MD 20746, USA. Email: filipfba@gmail.com; phone: +1 862-371-7266. I thank V.V. Chari, Kyle Herkenhoff, Michael Peters, Larry Warren, Jackie Zhao, and participants at CSU Fullerton and the Boston University Junior Macroeconomics Conference. All errors are my own. Any opinions and conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau. This paper does not use any confidential Census Bureau data.

# 1 Introduction

The dramatic rise in healthcare expenditures as a share of consumption over the last few decades has attracted attention, and some concern, from both policymakers and academics. US healthcare expenditures rose from 5 percent of GDP in 1960 to 18 percent in 2024 (Centers for Medicare & Medicaid Services 2026a). There is also substantial variation across countries—in contrast to the high expenditures in the US, spending in OECD countries averaged only 9.3 percent of GDP in 2024 (OECD 2025c). It is unclear if these variations are a natural consequence of rising incomes, driven by different rates of technological progress between healthcare and the rest of the economy, or affected by other factors, and these different explanations have different consequences for welfare.

This paper addresses a key challenge in understanding these trends, namely the tension between macro trends and micro data: the cross-section of healthcare expenditures across countries and time series within individual countries are hard to rationalize without an income elasticity of healthcare spending above one, yet this seems at odds with healthcare expenditures along the income distribution and with quasi-experimental evidence. As Eslami (2019) documents, healthcare expenditures are relatively flat across individual income levels, which seems inconsistent with an income elasticity much greater than zero, let alone one. Acemoglu et al. (2013) provide quasi-experimental evidence that healthcare expenditures rise less than one-for-one with incomes after an oil shock[1], which seems to bound the income elasticity below one.

I make three contributions to this debate. First, I adapt a model of structural transformation to the healthcare sector in order to understand the causes and consequences of rising healthcare spending. I use nonhomothetic preferences that allow both income and relative prices to affect expenditure shares. Similar to the aggregation logic in Getzen (2000, 2006), I represent pooled financing through a "health planner" that allocates resources across individuals before they divide their assigned expenditures

[1]More specifically, they use preexisting oil reserves interacted with contemporaneous oil prices as an instrument for income and infer an elasticity of only 0.7.

between healthcare and non-healthcare. This captures a key institutional detail: even in the US, individuals do not generally finance healthcare out of their own budgets. In most rich countries healthcare is chiefly financed by the government, and in the US it is financed by a mix of government spending, highly regulated private insurers that face strict limits on differentiated plans, and uncompensated hospital care. Out-of-pocket expenses account for only 11 percent of US healthcare expenditures (Centers for Medicare & Medicaid Services, 2026b). Healthcare expenditures can therefore be flat across individual income levels even when aggregate income effects are strong. The model also accommodates an aging population, a potentially important confounder because older individuals spend more on healthcare, richer countries tend to be older, and most countries have aged over time.

Second, I discipline the model using indirect inference by generating simulated data from the model to match panel data on healthcare expenditures as a share of consumption from OECD (2025a) and OECD (2025b) for 38 countries spanning half a century. Additionally, I match quasi-experimental evidence from Acemoglu et al. (2013) about the response of healthcare spending to a productivity shock outside the healthcare sector and data from Papanicolas et al. (2020) on the differences in health expenditures across ages in a sample of OECD countries in one specific year. I find that log healthcare expenditures are essentially linear in log overall consumption across the entire panel dataset, with an observed income elasticity well above one (see Figure 1). In this sample and specification, US observations generally lie only marginally above the fitted relationship. This extremely tight relationship leads the model to infer that income effects are the main force driving variation in healthcare spending in the time series and cross-section.

The quasi-experimental evidence of Acemoglu et al. (2013) does not contradict this because a productivity shock outside healthcare raises the relative price of healthcare and generates price effects as well as income effects, hence it commingles a price elasticity with the income elasticity. Within the model, matching the quasi-experimental response requires healthcare demand to have an own-price elasticity greater than one

in absolute value. A positive productivity shock outside healthcare raises both income and the relative price of healthcare. The income effect pushes healthcare spending upward, while the relative-price effect pushes its expenditure share downward. The observed less-than-one-for-one spending response therefore disciplines the strength of the price channel. Dodini et al. (2025) find that wages for doctors rose after a similar oil shock in Norway, consistent with the model's predictions of a relative price increase in healthcare after a positive productivity shock.

The model infers that aging only makes a small positive contribution, as the spending differences in Papanicolas et al. (2020) scaled by differences in the age distribution across time and place are small relative to overall differences in healthcare spending. It also infers that differences in the rate of technological growth in healthcare relative to the rest of the economy have only a small effect because the slope of healthcare spending with respect to time in the panel is small after controlling for overall consumption. To match the slight upward trend in healthcare spending in the panel, the model infers that the relative productivity of healthcare must be rising slightly over time and the relative price must be falling[2], which generates price effects that work in the opposite direction of the non-healthcare productivity shock in Acemoglu et al. (2013). An interesting consequence of this parameterization is that increasing the relative productivity of healthcare will *increase* its share of spending. This finding is a cautionary note for policymakers looking to control healthcare spending.

In a third contribution, I use the calibrated model to perform a series of structural decomposition exercises and counterfactuals to understand the determinants of healthcare spending in the past, present, and future, and find a dominant role for income effects in all three. First, I decompose differences in healthcare spending across time and place into the relative effects of rising incomes, aging, and differences in the rate of technological progress. I find that incomes play the dominant role, with aging explaining a small part of the variation and relative price effects explaining al-

[2]I avoid using healthcare price indices because credibly measuring *quality-adjusted* price growth in healthcare is an ongoing challenge. Dunn et al. (2022) get estimates of healthcare price growth ranging from 0.77 percent above inflation to 3.1 percent below it under different assumptions of the value of a statistical life year, holding the rest of their analysis fixed.

most none. Second, I focus on the most recent cross-section of data and ask how much healthcare spending across countries would converge if their age structures or productivity levels were equalized. As with the panel, the cross-sectional variation is almost entirely due to income differences. Finally, I use the model to simulate the growth of healthcare expenditures in the future and find that the healthcare share of expenditures is most sensitive to overall income growth.

## 1.1 Literature Review

This paper contributes to a burgeoning literature that tries to explain the rise in healthcare expenditures as a share of GDP.

One strand of this literature emphasizes income as a driver of health spending (Newhouse, 1977; Getzen, 2000, 2006). Getzen's aggregation argument is especially close to the mechanism here: because care is largely financed collectively, group income rather than a patient's own income can determine spending, reconciling near-zero individual income elasticities with national elasticities above one. Random Critical Analysis (2023) provides a particularly clear exposition of this aggregate-income view and provides direct evidence based on actual individual consumption (AIC), which I also use in this paper. Hall and Jones (2007) explain why rising income can naturally raise health spending, while Chandra and Skinner (2012) and Cutler (2018) emphasize technology, value, and inefficiency in assessing that growth. Eslami (2019) documents that healthcare expenditures across income quintiles are flat or slightly decreasing and explains why correlated earnings and health risks complicate estimates from within-country cross-sectional variation. My contribution is to embed the pooled-financing insight in a structural-transformation framework that separates income and relative-price effects and calibrate it to both aggregate and quasi-experimental evidence.

Another strand of the literature has focused on technological progress in the healthcare sector. Jones (2002) argues that medical advances can rationalize the rising share of spending going to healthcare. Some formerly incurable diseases (i.e. ones with

an infinite price for a cure) can be treated today, and Jones (2002) argues that it is reasonable that spending should shift towards healthcare as such prices fall. Conversely, Anderson et al. (2003) argue that the US's high spending is driven by its high quality-adjusted relative prices—that is, the US spends more on healthcare because of its higher prices. Fonseca et al. (2021) use a rich quantitative life-cycle model featuring factors like smoking, obesity, insurance, tax and transfer schemes, and overall income growth to infer that technological progress in healthcare explains half the rise in life expectancy from 1965-2005. Fonseca et al. (2023) also use a detailed structural lifecycle model to study cross-country differences in healthcare expenditures. My model provides a complementary approach to estimating this rate of technological progress with a simple structural transformation model. It allows me to infer that aggregate productivity growth in healthcare is broadly similar to, if not slightly faster than, growth in the rest of the economy.

This paper also contributes to the literature on structural transformation, which has so far focused on the transition away from agriculture into manufacturing and then into services. Boppart (2014) reintroduces price-independent generalized linear (PIGL) preferences, which are nonhomothetic and capture both income and relative price effects, and uses them to decompose the shift away from goods and into services into an income effect and a relative price effect. This paper kicked off a growing literature using nonhomothetic preferences to study structural transformation. Eckert and Peters (2025) use such preferences to study the industrialization of rural America, and Fan et al. (2023) use them to study the increased share of spending on services in India. Comin et al. (2021) argue that nonhomothetic CES preferences are better able to match key facts about structural transformation from agriculture to manufacturing and then services. I contribute by studying the shift towards one major category of services, namely healthcare, and adapting the structural transformation machinery to key institutional details of the "market" for healthcare.

This paper contributes to the literature on differences in technological progress between healthcare and the rest of the economy. There is still no consensus even on the

sign of this difference. Dunn et al. (2022), Dauda et al. (2020), Jones (2002), and Anderson et al. (2003) all use different methods and arrive at different conclusions about whether quality-adjusted prices are rising faster or slower in healthcare relative to the rest of the economy. Aside from the inherent difficulty of adjusting for quality, these estimates are also highly sensitive to the value of a statistical life year. My estimates are complementary to this literature as I use a completely different methodology and estimate relative productivity growth using a structural model matched to aggregate moments and quasi-experimental evidence. I find that relative productivity in healthcare is rising slightly faster than in the rest of the economy.

Finally, another branch of the literature has bridged models of growth and healthcare to understand the long-run dynamics of the healthcare sector. Jones (2016) studies the asymptotic allocation of resources to life extension. Pretnar and Feldman (2026) argue that markup growth has inflated relative prices in healthcare, and also find that population aging makes only a minor contribution to the rising health expenditure share. Hejkal et al. (2025) develop a model of endogenous technological change in healthcare and use it to study the evolution of mortality across place and time. Huetsch et al. (2026) propose a unified theory of endogenous technological progress across healthcare and the rest of the economy that seeks to explain long-run patterns of income, healthcare expenditures, and life expectancy. My focus is on variation in the healthcare expenditure share across time and space, which differs from the remaining papers in the literature. I also contribute by using quasi-experimental evidence rather than price data to discipline technological progress in healthcare relative to the rest of the economy.

## 1.2 Roadmap

The paper proceeds as follows: In Section 2, I outline the model. In Section 3, I introduce my key data sources and use them to discipline the model. In Section 4, I perform a series of exercises with the calibrated model. In Section 5, I offer some thoughts on the results and on room for future research. Section 6 concludes.

# 2 Model

I now introduce the structural model, which allows me to bridge theory and evidence to conduct counterfactual exercises. First I describe a simplified baseline model, then I extend the model to include multiple regions that can trade with each other and have different productivities in each sector, allowing me to connect to the evidence in Acemoglu et al. (2013).

## 2.1 Baseline

I consider a closed economy with a measure one of individuals who belong to one of three age categories: 0-19 years old, 20-64 years old, and 65+ years old, which I denote as $a \in \{\text{child}, \text{adult}, \text{old}\}$. Time $t$ is discrete. The share of each age group in a given period is $\pi_{a,t}$, with $\sum_a \pi_{a,t} = 1$. Individuals have preferences over healthcare $h$ and a composite non-healthcare good $n$. I denote $x_{a,t}$ as total nominal expenditures (with the non-healthcare good as the numéraire), and $p_{h,t}$ as the (relative) price of healthcare. Each individual supplies a single unit of labor inelastically to either the healthcare or non-healthcare sector, both of which have a linear labor-only technology. Labor $l$ is

perfectly mobile across the two sectors, which have productivities $z_{s,t}$ for $s \in \{h, n\}$. Output per worker[3] in each sector is therefore given by:

$$y_{s,t} = z_{s,t} l_{s,t}, \qquad s \in \{h, n\}. \tag{1}$$

Wages $w_{s,t}$ for $s \in \{h, n\}$ are set competitively, as are prices for each good. Hence, relative prices $p_{h,t}$ are the inverse of relative productivities:

$$p_{h,t} = \frac{z_{n,t}}{z_{h,t}}. \tag{2}$$

There is no savings technology, so each time period can be treated as a separate static economy. I therefore drop time subscripts from here on out. Individuals in group

[3]Note that differences in hours per worker across time and space are absorbed into the productivity terms $z_{s,t}$.

$a$ have nonhomothetic preferences[4] over the two goods, with curvature parameters $\rho_h, \rho_n \in (0,1)$ and productivity parameters $\kappa_a > 0$ that differ across age groups:

$$U_a(h,n) = \log\left(\left(\kappa_a h^{\rho_h} + n^{\rho_n}\right)^{1/\rho_h}\right). \tag{3}$$

This preference specification is closely related to the familiar constant elasticity of substitution utility function—if the $\rho$ parameters were the same, then $1/(1-\rho)$ would be the elasticity of substitution. Allowing them to differ allows for more flexible income and price effects, which is important for matching the data.

The indirect utility of an agent with budget $x$ facing price $p_h$ can be written as:

$$V_a(x, p_h) = \max_{h,n} U_a(h,n) \mid p_h h + n \leq x. \tag{4}$$

The indirect utility function in Equation 4 does not admit a closed-form solution for expenditure shares, but solving the model is straightforward. The intratemporal allocation is characterized by the ratio of marginal utilities,

$$\frac{u_{a,h}(h,n)}{u_{a,n}(h,n)} = \frac{\rho_h}{\rho_n}\kappa_a \frac{h^{\rho_h - 1}}{n^{\rho_n - 1}} = p_h. \tag{5}$$

The last equality in Equation 5 follows from the first-order condition for the consumer's problem, which states that the marginal rate of substitution must equal the relative price. Combining this condition with the budget constraint yields an implicit demand equation. Define age group $a$'s healthcare expenditure share as

$$s_{h,a} \equiv \frac{p_h h}{x}. \tag{6}$$

[4]I do not use price-independent generalized linear (PIGL) preferences as in Boppart (2014) because I was unable to find a calibration that kept expenditure shares between 0 and 1 for all age groups in all countries in all years while still matching the key data moments. Nonhomothetic CES preferences as in Comin et al. (2021) guarantee an expenditure share between 0 and 1, but if the two goods are substitutes then marginal utility is not necessarily strictly diminishing in expenditure, which complicates the problem of the health planner as equalizing marginal utilities across age groups does not necessarily attain the optimal solution. Stone-Geary preferences force the nonhomotheticity of preferences to disappear over time, and they produce a corner solution for expenditure shares for luxury goods at low income levels. The downside of my preference specification is that it does not provide analytic expenditure shares.

Substituting $h = s_{h,a}x/p_h$ and $n = (1 - s_{h,a})x$ into Equation 5 gives

$$p_h^{\rho_h} = \frac{\rho_h \kappa_a}{\rho_n} x^{\rho_h - \rho_n} s_{h,a}^{\rho_h - 1} (1 - s_{h,a})^{1-\rho_n}. \tag{7}$$

Although Equation 7 does not generally admit a closed-form solution for $s_{h,a}$, it makes the relevant comparative statics transparent. Implicit differentiation gives

$$\frac{\partial s_{h,a}}{\partial \log p_h} = -\frac{\rho_h}{\frac{1-\rho_h}{s_{h,a}} + \frac{1-\rho_n}{1-s_{h,a}}} < 0, \tag{8}$$

$$\frac{\partial s_{h,a}}{\partial \log x} = \frac{\rho_h - \rho_n}{\frac{1-\rho_h}{s_{h,a}} + \frac{1-\rho_n}{1-s_{h,a}}}. \tag{9}$$

Thus, when $\rho_h, \rho_n \in (0, 1)$, an increase in the relative price of healthcare lowers its expenditure share. Since $h = s_{h,a}x/p_h$, the Marshallian own-price elasticity of healthcare demand, holding total expenditure fixed, is

$$\frac{\partial \log h}{\partial \log p_h} = \frac{1}{s_{h,a}} \frac{\partial s_{h,a}}{\partial \log p_h} - 1 < -1. \tag{10}$$

Healthcare demand therefore has an own-price elasticity greater than one in absolute value under the maintained restriction $\rho_h, \rho_n \in (0, 1)$. Equation 9 also shows that the healthcare expenditure share rises with total expenditure when $\rho_h > \rho_n$. Intuitively, in that case marginal utility diminishes more slowly with healthcare consumption than with non-healthcare consumption, making healthcare a luxury good. Note that setting $\rho_h, \rho_n < 0$ is also mathematically valid, but it would not be consistent with the data.

The $\kappa$ terms capture differences in relative productivity by age group—that is, in the empirically-relevant region of the parameter space where the $\kappa$ terms are strictly increasing[5] in age, healthcare is more productive (i.e. has a lower quality-adjusted price) for the old than for adults, and more productive for adults than for the young. To understand why healthcare could plausibly be relatively more productive for the

[5] I impose no restrictions on $\kappa$ but find that the parameters that maximize fit are strictly increasing in age.

old than the young, consider the case of COVID vaccines and cancer screenings. It takes many more vaccines and screenings to prevent a death among young people than among old people—see Goldstein et al. (2021) and Duffy et al. (2010), respectively.

Importantly, an individual's budget $x$ is *not* their own labor income. Similar to the pooled-financing interpretation in Getzen (2000, 2006), the "health planner" is a tractable stand-in for the insurers and government programs that account for the majority of health expenditures. It redistributes among agents to maximize utility. I assume that the health planner reallocates expenditures in "dollar" terms across individuals and that individuals then decide how to split their expenditures across the two goods. The planner takes as given the population shares $\pi_a$ of each age group and the relative price of healthcare $p_h$, and it chooses the nominal expenditure $x_a$ (in units of the non-healthcare numéraire) assigned to each age group $a$ to maximize population-weighted indirect utility, $\sum_a \pi_a V_a(x_a, p_h)$, subject to the constraint that average assigned expenditure cannot exceed the aggregate budget, $\sum_a \pi_a x_a \leq x$. Each age group $a$ then takes its assigned $x_a$ and solves the individual problem in Equation 4, dividing $x_a$ between healthcare $h$ and the non-healthcare good $n$. This assumption means that the problem of a planner with aggregate budget $x$ can be written as:

$$\max_{\{x_a\}} \sum_a \pi_a V_a(x_a, p_h) \mid \sum_a \pi_a x_a \leq x. \tag{11}$$

The health planner is a reasonable abstraction given the extent of government intervention in healthcare in most countries. Many countries have state-run healthcare programs, and even the US has programs like Medicare, Medicaid, and the Affordable Care Act. As documented in Coughlin et al. (2021), hospitals in the US also provide tens of billions of dollars of uncompensated care to patients who have no realistic way to pay at all. Bassler (2024) also describes institutional barriers limiting the ability of employers to offer more generous health plans to their higher-paid employees. It is therefore not realistic to treat healthcare expenditures as coming out of an individual's own budget, even in the US. The health planner is a reduced-form stand-in for the mix of government spending, subsidized and regulated private insurers, and uncompen-

sated care provided by hospitals. These components actually account for the majority of healthcare expenditures in the US and in most other rich countries. Because the planner represents these pooled financing arrangements in reduced form, the model does not separately identify public-insurance outlays, private-insurance outlays, or uncompensated care, nor their growth over time. Separating these components would require additional structure, such as distinct financing wedges or insurer budget constraints, and additional moments to discipline them.

As money is fungible, this is equivalent to purchasing the healthcare good directly and allocating it across individuals based on age, then having individuals choose their optimal health expenditures. As noted earlier, my preference specification does not allow me to calculate expenditure shares analytically. I therefore solve numerically by breaking the problem into two steps: first, I guess the Lagrange multiplier on the budget constraint in the planner's problem, then I solve for the level of $x_a$ such that each group's marginal utility is equal to this multiplier. I then search for the multiplier such that the expenditures $x_a$ respect the budget constraint $\sum_a \pi_a x_a \leq x$. This reduces the problem to a set of nested one-dimensional monotone root-finding problems.

**Equilibrium in the baseline economy.** An equilibrium is a collection of relative prices $(p_h, w)$, labor allocations $(l_h, l_n)$, outputs $(y_h, y_n)$, age-specific expenditures $\{x_a\}_{a\in\mathcal{A}}$, and age-specific consumption allocations $\{h_a, n_a\}_{a\in\mathcal{A}}$, with $\mathcal{A} = \{\text{child}, \text{adult}, \text{old}\}$, such that:

1. Firms maximize profits under the linear technology[6] in Equation 1.
2. Given prices and assigned expenditure $x_a$, each age group solves the indirect utility problem in Equation 4 (with $(x, p_h)$ replaced by $(x_a, p_h)$).
3. The health planner solves the problem in Equation 11, with aggregate budget $x = p_h y_h + y_n$.

[6]With competitive markets, $p_s z_s \leq w$, with equality if $l_s > 0$. If both sectors are active, relative prices satisfy Equation 2.

4. Markets clear:

$$l_h + l_n = 1, \qquad y_h = \sum_{a \in \mathcal{A}} \pi_a h_a, \qquad y_n = \sum_{a \in \mathcal{A}} \pi_a n_a. \tag{12}$$

## 2.2 Extended Model with Trade

I now introduce a stylized spatial extension of my model, with multiple regions inside a country.[7] I drop time subscripts in this section for clarity.

There are $N$ regions within a country denoted by $r$. Each region is endowed with one inelastically supplied unit of labor that it can allocate between the production of a local healthcare good $y_{h,r}$ and the production of a local non-healthcare good $y_{n,r}$. As before, labor $l$ is perfectly mobile between the two sectors $s$, and production is linear with sector-region specific productivities $z_{s,r}$, hence:

$$y_{s,r} = z_{s,r} l_{s,r}. \tag{13}$$

Each regionally-produced variety can be consumed locally or exported to one of the other regions. When consumed locally I denote it as $y_{s,rr}$, and when exported I denote it as $y_{s,rj}$. As noted earlier, the price of a domestically-consumed good is $p_{s,rr}$, and $p_{s,rj}$ denotes the delivered price paid by consumers in destination region $j$ for the sector-$s$ variety produced in origin region $r$.

In each region, the final healthcare and non-healthcare goods are CES aggregators of all regional varieties. Regions are arranged on the spokes of a hub-and-spoke trade network, hence all are a distance of $d$ away from each other. Trade is subject to the usual iceberg costs, with a distance elasticity $\theta_s$ in each sector. In particular, if a firm ships $q$ units of a sector-$s$ variety from region $r$ to region $j$, only $d^{-\theta_s} q$ units arrive. Equivalently, delivering one unit requires shipping $d^{\theta_s}$ units. Thus, $d^{-\theta_s}$ is the fraction that arrives, and $1 - d^{-\theta_s}$ is the fraction that melts away. A higher value of $\theta_h$ relative

[7] Cross-country evidence helps me discipline the income effect; regional responses to the quasi-experiment in Acemoglu et al. (2013) allow me to calibrate substitution effects by generating relative price effects.

to $\theta_n$ captures the idea that healthcare is harder to trade than most other sectors.

Suppressing the usual discrete-CES normalization constant, the sector-$s$ composite in destination region $r$ is

$$Q_{s,r} = \left[ y_{s,rr}^{\frac{\sigma-1}{\sigma}} + \sum_{j \neq r} \left( d^{-\theta_s} y_{s,jr} \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \tag{14}$$

where $\sigma > 1$ is the elasticity of substitution across regional varieties. Because $p_{s,jr}$ is the delivered price paid by consumers in destination region $r$, the corresponding CES price index is

$$P_{s,r} = \left[ p_{s,rr}^{1-\sigma} + \sum_{j \neq r} p_{s,jr}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \tag{15}$$

In each destination $r$, a competitive final-good producer assembles the sector-$s$ composite $Q_{s,r}$ by minimizing expenditure on delivered regional varieties, $\sum_j p_{s,jr}\, q_{s,jr}$, subject to the CES technology in Equation 14, where $q_{s,rr} = y_{s,rr}$ and $q_{s,jr} = d^{-\theta_s} y_{s,jr}$ for $j \neq r$ denote the quantities that actually arrive in region $r$. Cost minimization yields the price index in Equation 15 together with the variety demands

$$q_{s,jr} = \left( \frac{p_{s,jr}}{P_{s,r}} \right)^{-\sigma} Q_{s,r}. \tag{16}$$

Equivalently, in terms of shipped quantities,

$$y_{s,rr} = \left( \frac{p_{s,rr}}{P_{s,r}} \right)^{-\sigma} Q_{s,r}, \qquad d^{-\theta_s} y_{s,jr} = \left( \frac{p_{s,jr}}{P_{s,r}} \right)^{-\sigma} Q_{s,r} \quad \text{for } j \neq r, \tag{17}$$

which determine the equilibrium quantity of shipments into region $r$.

Regional income is taxed away at rate $\tau$ and then rebated back lump sum. This captures the concerns raised in Random Critical Analysis (2023) that the Acemoglu et al. (2013) quasi-experiment may understate the income effect if some income is taxed away.

A region's pretax income is the sum of all its output across all sectors, exported and consumed domestically:

$$Y_r = \sum_s \left( p_{s,rr} y_{s,rr} + \sum_{j \neq r} p_{s,rj} d^{-\theta_s} y_{s,rj} \right). \tag{18}$$

Defining $\bar{Y}$ as mean income across regions, region $r$'s posttax income is therefore:

$$Y_{r,p} = (1 - \tau) Y_r + \tau \bar{Y}. \tag{19}$$

I assume that labor is perfectly mobile across sectors, that the output of each region is transferable between use at home and exporting, and that markets for both workers and goods are competitive. This last assumption implies that each regional variety is produced by a unit measure of identical firms and that the price of each variety is equal to its marginal cost. Because there is no local monopolist, there is no markup for the goods. Local prices are therefore equal to marginal costs, which are the wages of the workers $w_r$:

$$p_{s,rr} = \frac{w_r}{z_{s,r}}. \tag{20}$$

Hence, the price ratios of the locally-consumed varieties are the inverse of their relative productivities:

$$\frac{p_{h,rr}}{p_{n,rr}} = \frac{z_{n,r}}{z_{h,r}}. \tag{21}$$

For exports, iceberg costs imply that the delivered price paid by consumers in destination region $j$ is

$$p_{s,rj} = d^{\theta_s} p_{s,rr} \qquad \text{for all } j \neq r. \tag{22}$$

Equivalently, producer revenue per unit shipped is $d^{-\theta_s} p_{s,rj} = p_{s,rr}$.

Individuals in each region again have the same indirect utility function, now with an explicit regional price for the non-healthcare good. Unlike in the baseline model, a single numéraire normalization can set $P_{n,r} = 1$ in one reference region, but it cannot set every $P_{n,r}$ to one when these prices differ across regions:

$$V_a(x, P_{h,r}, P_{n,r}) = \max_{h,n} U_a(h, n) \mid P_{h,r} h + P_{n,r} n \leq x. \tag{23}$$

The health planner, now specific to each region, again equates marginal utility across age groups.

In most of the quantitative exercises, I treat all regions as symmetrical, in which case this version of the model reduces (after a few renormalizations) to the baseline. The only exception is that I shock $z_n$ in one region to replicate the Acemoglu et al. (2013) quasi-experiment. I use the non-healthcare good in the non-shocked regions as the numéraire in that case.

**Equilibrium in the spatial economy.** An equilibrium is a collection of regional wages $\{w_r\}$, delivered prices $\{p_{s,rj}\}$, price indices $\{P_{s,r}\}$, labor allocations $\{l_{s,r}\}$, shipments $\{y_{s,rj}\}$, composite quantities $\{Q_{s,r}\}$, incomes $\{Y_r, Y_{r,p}\}$, age-specific expenditures $\{x_{a,r}\}$, and age-specific consumption allocations $\{h_{a,r}, n_{a,r}\}$, such that:

1. Firms maximize profits under the linear technology[8] in Equation 13, with total output split across destinations: $y_{s,r} = \sum_{j=1}^{N} y_{s,rj}$.

2. In each $r$, the final-good producer's cost minimization yields the sectoral composites in Equation 14, the price index in Equation 15, and the variety demands in Equations 16–17, which determine the shipments $\{y_{s,jr}\}$ into region $r$.

3. Regional pretax income is given by Equation 18 and post-tax income by Equation 19, with $\bar{Y} = \frac{1}{N}\sum_{j=1}^{N} Y_j$.

4. Given $(P_{h,r}, P_{n,r})$ and $x_{a,r}$, each age group solves the indirect utility problem in Equation 23 (with $x$ replaced by $x_{a,r}$).

5. The regional health planner solves the analogue of Equation 11 for region $r$:

$$\max_{\{x_{a,r}\}_{a\in\mathcal{A}}} \sum_{a\in\mathcal{A}} \pi_{a,r} V_a(x_{a,r}, P_{h,r}, P_{n,r}) \quad \text{s.t.} \quad \sum_{a\in\mathcal{A}} \pi_{a,r} x_{a,r} \leq Y_{r,p}. \tag{24}$$

[8] Competitive pricing implies $p_{s,rr} z_{s,r} \leq w_r$, with equality if $l_{s,r} > 0$, and delivered prices for $j \neq r$ satisfy Equation 22.

6. Markets clear:

$$\sum_{s\in\{h,n\}} l_{s,r} = 1, \tag{25}$$

$$Q_{h,r} = \sum_{a\in\mathcal{A}} \pi_{a,r} h_{a,r}, \qquad Q_{n,r} = \sum_{a\in\mathcal{A}} \pi_{a,r} n_{a,r}. \tag{26}$$

# 3 Calibration

In this section, I bridge the model and data by matching a set of key facts that are informative about the different forces in the model, with the goal of using this calibrated model to make quantitative statements about the past, present, and future of healthcare spending. I discipline the model using data on healthcare expenditures and consumption from OECD (2025a) and OECD (2025b), as well as quasi-experimental evidence from Acemoglu et al. (2013) and data on healthcare expenditures across ages from Papanicolas et al. (2020).

## 3.1 Healthcare and Consumption Data

I begin by documenting the key facts about healthcare expenditures and consumption that I use to discipline the model.

I get health expenditures and consumption from OECD (2025a) and OECD (2025b) for an unbalanced panel of 38 countries starting in 1970 and running until 2024. As in Random Critical Analysis (2023), I use actual individual consumption (AIC) as my measure of material well-being. This includes all household consumption, expenditures by non-profit institutions serving households, and government consumption expenditures on goods and services that are directly consumed by households (Eurostat, 2026). For my baseline measure of healthcare expenditures, I use the System of Health Accounts (SHA) measure of healthcare expenditures (Eurostat et al., 2017), which includes all expenditures on healthcare goods and services, including those paid for by government programs and private insurers. The SHA measure is the most comprehensive and longest-running measure of healthcare expenditures available, and it is

the one used in the majority of the literature on healthcare expenditures. However, it does not perfectly align with AIC, as it includes collective health services such as epidemiological surveillance and administration. I therefore perform a robustness exercise using a measure of healthcare expenditures that is entirely nested in AIC, which I describe in more detail below.

There is an extremely tight relationship between AIC and healthcare expenditures in both the time series and cross-section, as shown in Figure 1. This figure includes all the countries and years in my data. The best-fit line (solid orange) is steeper than the slope-one line (dashed green, running through the median data point), indicating an income elasticity above one. The red squares are the US time series, and generally are only marginally above the best-fit line.

I perform a robustness exercise for the smaller set of country-years where the OECD breaks out AIC into its components, including a measure of healthcare expenditures entirely nested in AIC. In addition, I also report the results after subtracting healthcare expenditures from AIC to get a measure of non-healthcare expenditure. This last exercise guards against the concern that the tight relationship in Figure 1 is mechanically driven by the fact that most healthcare expenditures are included in AIC. I run regressions of the form:

$$\log(HC_{i,t}) = \alpha + \beta \log(AIC_{i,t}) + \gamma t + \epsilon_{i,t} \tag{27}$$

and report the output in Table 1. The results are robust to the choice of healthcare expenditure measure and to the choice of consumption measure, with the coefficient on AIC always above 1 and significant at the 1% level. If anything, the net-of-healthcare-consumption regressions have higher coefficients on AIC, which suggests that the tight relationship in Figure 1 is not mechanically driven by the fact that most healthcare expenditures are included in AIC.

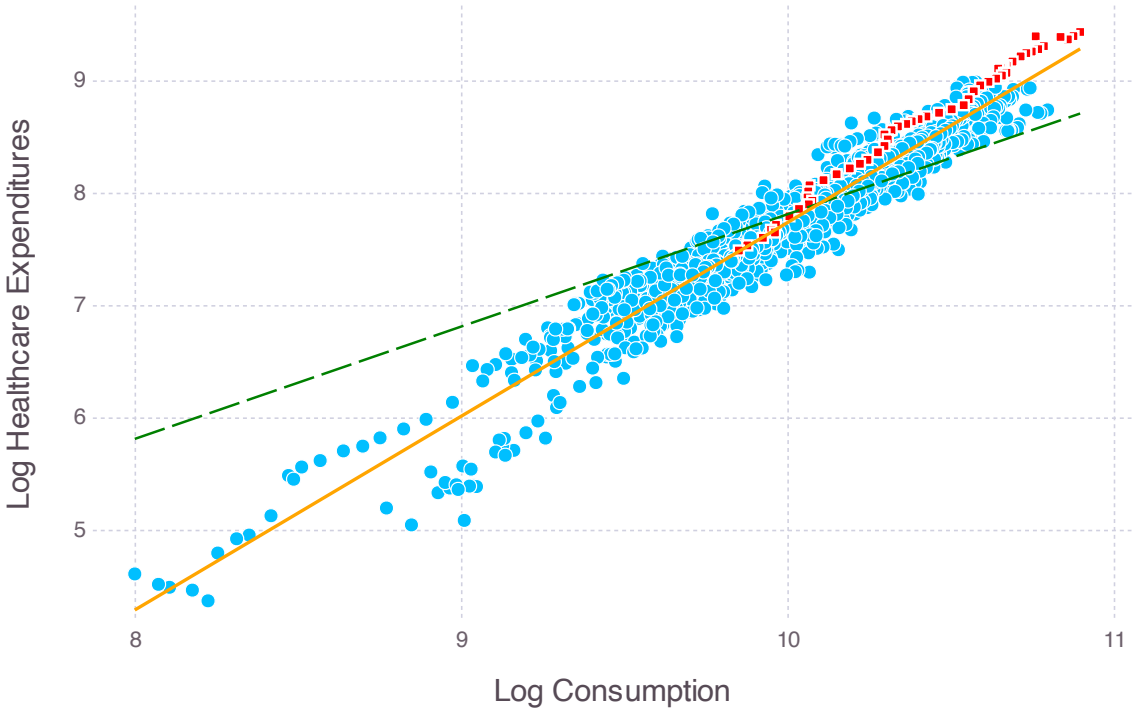



Figure 1: Cross-Sectional Healthcare Expenditures vs Consumption

*Notes*: This figure reports the log of actual individual consumption on the horizontal axis and health spending on the vertical axis across all countries and years in the OECD data. The orange solid line is the best fit line, and the green dashed line is the slope-one line running through the median data point. The red squares correspond to the US across years. All observations are in purchasing power parity adjusted constant dollars.

## 3.2 Externally Calibrated Parameters

I assume that the technological frontiers for healthcare and non-healthcare grow at constant rates $g_h, g_n$. Countries have the same relative productivities in each time period but differ in their levels of technology $z_{i,t}$. This is a reasonable assumption for the relatively rich OECD countries in this sample, as differences in their GDP per capita are mostly mediated by hours worked (Fernández-Villaverde et al. 2025), which are isomorphic to level shifts in $z$. That is, differences in hours worked are equivalent to scaling each productivity up or down by a common factor. The initial ratio of productivity in healthcare vs non-healthcare is $r_0$. Hence, in country $i$ at time $t$:

$$z_{n,i,t} = z_{i,t}, \quad z_{h,i,t} = z_{i,t} r_0 e^{(g_h - g_n)t}. \tag{28}$$

Table 1: Robustness to Alternative Health and Consumption Measures

| | SHA | | A0106 | |
|---|---|---|---|---|
| | (1) log(AIC) | (2) log(AIC−HC) | (3) log(AIC) | (4) log(AIC−HC) |
| (Intercept) | -15.04*** | -18.46*** | -15.07*** | -18.1*** |
| | (2.166) | (2.463) | (3.738) | (4.194) |
| Consumption | 1.664*** | 1.748*** | 1.532*** | 1.588*** |
| | (0.0877) | (0.1085) | (0.09923) | (0.1237) |
| Year $t$ | 0.003075** | 0.004467*** | 0.003693* | 0.005013** |
| | (0.001379) | (0.001587) | (0.002238) | (0.002534) |
| $N$ | 1,633 | 1,633 | 292 | 292 |
| $R^2$ | 0.9392 | 0.9131 | 0.8891 | 0.8428 |

*Notes:* This table reports OLS estimates of $\log(\text{HC}_{i,t})$ on a consumption measure and a year trend. Columns (1)–(2) use the SHA health expenditure measure; columns (3)–(4) use the A0106 (National Accounts) health expenditure measure. Columns (1) and (3) use $\log(\text{AIC}_{i,t})$ as the consumption measure; columns (2) and (4) subtract healthcare expenditures from AIC and use $\log(\text{AIC}_{i,t} - \text{HC}_{i,t})$. I cluster standard errors at the country level.

Significance levels: * p<0.1, ** p<0.05, *** p<0.01.

Because I treat $n$ as the numéraire, measured consumption will be proportional to $z_{i,t}$. The levels $z_{i,t}$, the initial ratio $r_0$, and the values of $\kappa$ all enter the health planner's problem multiplicatively. I therefore normalize $z_{i,t}$ to be equal to AIC and $r_0$ to be equal to 1. Additionally, because I have country-by-year technology shifters, only the difference in the rate of technological growth $g_h - g_n$ is separately identifiable.

Relative prices between healthcare and non-healthcare therefore evolve as follows:

$$\frac{p_{h,i,t}}{p_{n,i,t}} = \frac{z_{n,i,t}}{z_{h,i,t}} = e^{(g_n - g_h)t}. \tag{29}$$

Dingel et al. (2026) report a healthcare trade elasticity with respect to distance of about -1.7, whereas Boulatoff et al. (2022) report a distance elasticity of -1.1 for trade in services across Canadian provinces. I set $\sigma$ to a standard value of 6 (common values range from 4 in Allen and Arkolakis 2014 to around 8 in Eckert and Peters 2025), and I set the two $\theta$ parameters so that the model-implied distance elasticities $(1-\sigma)\theta$ match the data.

The units of geography in Acemoglu et al. (2013) are 247 Economic Sub Regions,

and the units in Dingel et al. (2026) are 306 Hospital Referral Regions. I set $N = 275$ to roughly split the difference.

Federal taxes as a share of GDP are fairly stable at 0.2 according to U.S. Office of Management and Budget (2023), hence I set $\tau = 0.2$.

## 3.3 Internally Calibrated Parameters

I still have $\rho_h$, $\rho_n$, $g_h - g_n$, $d$, and three values of $\kappa_a$ to solve for. Here, $\rho_h$ and $\rho_n$ govern the curvature of utility in healthcare and non-healthcare, $g_h - g_n$ controls the trend in relative prices over time, $d$ controls how hard it is to trade healthcare across regions (and therefore regional expenditure smoothing), and the age-specific $\kappa_a$ terms govern the effective productivity of healthcare by age. To calibrate these parameters, I use indirect inference and compare model-generated and real data. I can simulate overall consumption and healthcare demand in each age group in a country-year in the panel if I know the productivity and age structure of the country in that year. I calibrate productivities externally as described in the last section, and I get data on population by age across countries and years from United Nations (2024). More specifically, I use the estimates of population in one-year age bins and then aggregate them up to match the age bins in Papanicolas et al. (2020). The model is exactly identified, with as many parameters as targets. As some parameters more

directly affect some target statistics, I can heuristically explain what fact disciplines each parameter.

I get three targets by summarizing Figure 1 in Equation 27. Intuitively, the slope on consumption in Equation 27 disciplines the gap between $\rho_h$ and $\rho_n$, the curvature parameters with respect to healthcare and non-healthcare. If the curvature parameter on healthcare is closer to 1 than non-healthcare, this implies that marginal utility with respect to healthcare diminishes more slowly, generating a higher elasticity of healthcare with respect to overall consumption. The slope on time pins down the contribution of relative price growth to healthcare spending over time, meaning it jointly affects $\rho_n$ and $g_h - g_n$ as these respectively control the response of healthcare spending to

Table 2: Results of Regression 27

| | $\log(HC_{i,t})$ | Predicted $\log(HC_{i,t})$ |
|---|---|---|
| (Intercept) | -15.04*** | -15.04*** |
| | (2.166) | (0.9843) |
| $\log(AIC_{i,t})$ | 1.664*** | 1.664*** |
| | (0.0877) | (0.0361) |
| Year $t$ | 0.003075** | 0.003075*** |
| | (0.001379) | (0.0006215) |
| $N$ | 1,633 | 1,633 |
| $R^2$ | 0.9392 | 0.9912 |

*Notes:* This table reports the results of running Equation 27 in both real and model-generated data. I cluster standard errors at the country level. Significance levels: * p<0.1, ** p<0.05, *** p<0.01.

relative prices and the speed of relative price changes. The constant in the regression pins down the level of the $\kappa$ terms, the age-specific healthcare productivity shifters. As shown in Table 2, the model fits the data extremely well.

I calibrate the $\kappa$ parameters, which govern the relative productivity of healthcare across age groups, by matching data from Papanicolas et al. (2020). The authors report that in a cross-section of eight OECD countries in 2015 the elderly spend about 3.4 times as much as adults, who in turn spend about 1.7 times as much as children. I set $\kappa_{\text{old}}$ and $\kappa_{\text{adult}}$ to match the mean ratios of the health spending of old people and adults to the health spending of the young across the eight countries.

Dingel et al. (2026) report that Medicare patients get about 20 percent of their treatment from providers in other regions. I set the distance parameter $d$, which governs how costly it is to ship the healthcare good across regions, to match this figure. As the import share in the case where all $N$ regions are symmetric is given by

$$s_h = \frac{(N-1)d^{\theta_h(1-\sigma)}}{1+(N-1)d^{\theta_h(1-\sigma)}} = \frac{N-1}{d^{\theta_h(\sigma-1)}+N-1}, \tag{30}$$

this implies that $d$ is approximately 61 for healthcare.

To discipline the response of healthcare spending to regional income shocks, I construct a counterfactual version of the Acemoglu et al. (2013) experiment inside the spatial model. The experiment is designed to isolate the effect of a productivity shock

outside healthcare that raises income in one region while also changing the relative price of healthcare. It does not identify income effects by itself in my model, as a change in relative prices violates the exclusion restriction. Acemoglu et al. (2013) find that a 1 percent income shock coming from oil (which I interpret as purely a change in $z_n$) leads to a 0.7 percent increase in health expenditures in the 1970-1990 time period. To match this, I shock $z_n$ in one specific region in the spatial model. This corresponds to a favorable oil shock in one part of the United States. As the regions are symmetrical before the productivity shock, the choice of which region is arbitrary. I choose the size of the non-health productivity shock so that the shocked region's revenue rises by 1 percent relative to a representative unshocked region, with revenue measured in units of the non-health good produced in an unshocked region. I then measure the implied change in log healthcare expenditures in the shocked region relative to that same unshocked region.

I normalize the price of the local non-health good in an unshocked region to one,

$$p_{n,jj} = 1. \tag{31}$$

Let $\delta$ denote the productivity shock to the non-health sector in the shocked region, so that

$$z_{n,r} = \delta z_{n,j}, \tag{32}$$

while healthcare productivity is held fixed. I solve jointly for $\delta$ and the local non-health price in the shocked region, $p_{n,rr}$. Given $(\delta, p_{n,rr})$, competitive pricing implies the wage ratio

$$\omega \equiv \frac{w_r}{w_j} = \delta p_{n,rr}. \tag{33}$$

This wage ratio pins down all local and delivered prices. In particular, the local healthcare price in the shocked region is

$$p_{h,rr} = \omega \frac{z_{n,j}}{z_{h,j}}, \tag{34}$$

while export prices are obtained by multiplying local prices by the appropriate iceberg trade cost. I then construct the CES price indices in the shocked and unshocked regions and solve the health planner's problem in each region to obtain total health and non-health expenditures.

Given these expenditures, I compute the producer revenue of the shocked region, $R_r(\delta, p_{n,rr})$, and the producer revenue of a representative unshocked region, $R_j(\delta, p_{n,rr})$, by summing sales across sectors and destinations using the CES expenditure shares. The equilibrium values of $\delta$ and $p_{n,rr}$ solve two equations. The first is goods-market clearing, which requires relative producer revenue to equal the wage ratio:

$$\log\left(\frac{R_r(\delta, p_{n,rr})}{R_j(\delta, p_{n,rr})}\right) - \log(\delta p_{n,rr}) = 0. \tag{35}$$

The second equation requires the shocked region's revenue to rise by 1 percent relative to a representative unshocked region, measured as a treated-minus-control change in logs:

$$\Big[\log R_r(\delta, p_{n,rr}) - \log R_r(1,1)\Big] - \Big[\log R_j(\delta, p_{n,rr}) - \log R_j(1,1)\Big] = \log(1.01). \tag{36}$$

I solve this two-equation system numerically. After finding the equilibrium pair $(\delta^\star, p^\star_{n,rr})$, I compute the model-implied analogue to the Acemoglu et al. (2013) regression coefficient as the ratio of the treated-minus-control change in log healthcare expenditures to the treated-minus-control change in log regional revenue:

$$\varepsilon^{AFN}_{h,R} = \frac{\Big[\log E_{h,r}(\delta^\star, p^\star_{n,rr}) - \log E_{h,r}(1,1)\Big] - \Big[\log E_{h,j}(\delta^\star, p^\star_{n,rr}) - \log E_{h,j}(1,1)\Big]}{\Big[\log R_r(\delta^\star, p^\star_{n,rr}) - \log R_r(1,1)\Big] - \Big[\log R_j(\delta^\star, p^\star_{n,rr}) - \log R_j(1,1)\Big]}, \tag{37}$$

where $E_{h,r}$ and $E_{h,j}$ are total healthcare expenditures in the shocked and representative unshocked regions. Differencing the shocked region against an unshocked control mirrors the comparison of more- and less-exposed areas in Acemoglu et al. (2013) and nets out any common, economy-wide movements induced by the shock. This pro-

cedure lets the model determine how much of the non-health productivity shock is passed through into local non-health prices versus local nominal income, while keeping the empirical counterfactual comparable to a 1 percent regional income shock. Because Equation 36 fixes the denominator at $\log(1.01)$, this elasticity is numerically equal to the model-implied treated-minus-control change in log healthcare expenditures divided by $\log(1.01)$.

In the calibrated model, Equation 37 evaluates to $\varepsilon_{h,R}^{AFN} = 0.70$, which closely matches the empirical target of 0.7 from Acemoglu et al. (2013).

This helps me target the difference between $\rho_h$ and $\rho_n$, which governs the strength of relative price effects. It also helps me pin down $g_h$, or more precisely the productivity gap. The baseline growth rate $g_n$ is not separable from the country-year productivity shocks $z_{i,t}$; hence I only target the difference $g_h - g_n$. Because a relative productivity increase in the non-healthcare sector leads to a decline in healthcare's share of expenditures, and because there is a slight positive time trend in the healthcare expenditure share in the panel, the model infers that $g_h > g_n$.

To help clarify how my model can rationalize the results in Acemoglu et al. (2013), I include a simplified directed acyclic graph (DAG) for the causal structure in their model and in mine in Figure 2. In the simplified DAG for Acemoglu et al. (2013) in Panel A, the shock affects health expenditures only through pretax income, which is the exclusion restriction that allows them to interpret the coefficient on income as an elasticity. In my model, shown in Panel B, a non-health productivity shock affects health expenditures through two channels: it changes pretax income, which maps into posttax income through the tax-and-rebate system, and it changes the relative price of health. Posttax income and the relative price jointly determine household demand and hence health expenditures. The presence of this second channel (borne out in a different country facing a similar shock in Dodini et al. 2025) means that the Acemoglu et al. (2013) quasi-experiment does not identify an income elasticity in my model, as a change in relative prices violates the exclusion restriction.

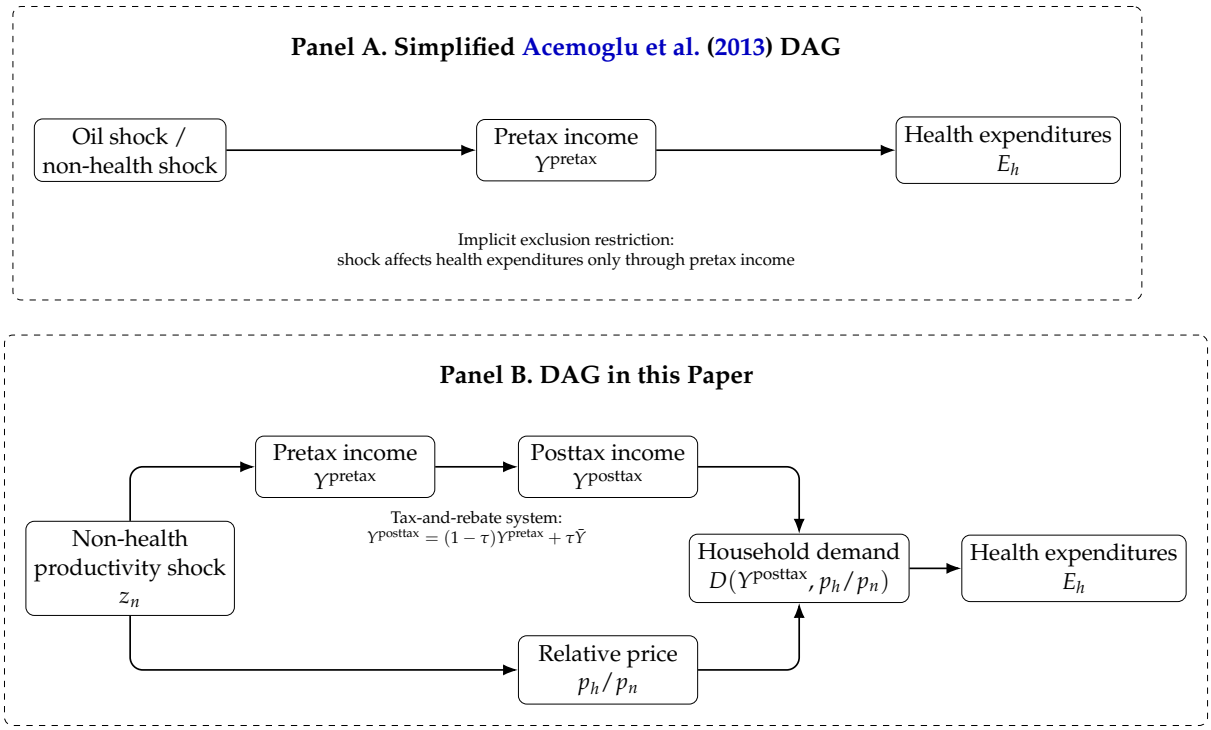



Figure 2: Directed Acyclic Graphs for Income Shocks and Health Expenditures

*Notes*: Panel A shows the simplified interpretation of Acemoglu et al. (2013), in which the shock affects health expenditures only through pretax income. Panel B shows the causal structure in my model. A non-health productivity shock changes pretax income and the relative price of health; posttax income and the relative price then jointly determine household demand.

Table 3: Calibration

| | Description | Value | Target | Data | Model |
|---|---|---|---|---|---|
| $\rho_h$ | Healthcare curvature | 0.44 | Consumption slope $\beta$ (jointly with $\rho_n$), Equation 27 | 1.664 | 1.664 |
| $\rho_n$ | Non-healthcare curvature | 0.08 | Time slope $\gamma$ (jointly with $\rho_h$), Equation 27 | 0.0031 | 0.0031 |
| $\kappa_{\text{child}}$ | Child productivity shifter | 0.0011 | Intercept, $\alpha$, in Equation 27 | -15.0 | -15.0 |
| $\kappa_{\text{adult}}$ | Adult productivity shifter | 0.0015 | Spending on adults/spending on children, PMLJ20 | 1.70 | 1.70 |
| $\kappa_{\text{old}}$ | Elderly productivity shifter | 0.0030 | Spending on elderly/spending on children, PMLJ20 | 5.73 | 5.73 |
| $g_h - g_n$ | Health productivity growth gap | 0.003 | Healthcare spending elasticity after oil shock, AFN13 | 0.70 | 0.70 |
| $d$ | "Distance" | 61.4 | Healthcare spending import share, DGLM26 | 20% | 20% |
| $\sigma$ | Elasticity of substitution | 6 | Standard | | |
| $\theta_n$ | Non-healthcare distance elasticity | 0.22 | BIK22 | | |
| $\theta_h$ | Health distance elasticity | 0.34 | DGLM26 | | |
| $\tau$ | Tax rate | 0.2 | U.S. Office of Management and Budget (2023) | | |
| $N$ | Number of regions | 275 | DGLM26, AFN13 | | |

*Notes:* This table reports the parameters for the calibrated model and compares model-generated moments with the data. All internally estimated parameters affect all moments, and the $\rho$ parameters in particular are jointly identified; I list parameters next to moments that are especially informative about them rather than implying a one-to-one mapping. The horizontal line divides the parameters that I calibrate internally (first group) and those I set externally (second group). AFN13 is Acemoglu et al. (2013), DGLM26 is Dingel et al. (2026), PMLJ20 is Papanicolas et al. (2020), and BIK22 is Boulatoff et al. (2022).

Table 4: Polynomial Fit: Data vs. Model

| | (Intercept) | $lcop_1$ | $lcop_2$ | $lcop_3$ | $lcop_4$ | Year $t$ | $N$ | $R^2$ |
|---|---|---|---|---|---|---|---|---|
| Data | -43.6*** | 23.22*** | -0.2634 | -0.2887 | 0.6903 | 0.02571*** | 1,633 | 0.9399 |
| | (1.781) | (1.151) | (0.6719) | (0.7767) | (1.022) | (0.0008866) | | |
| Model | -43.6*** | 23.22*** | -0.9769*** | -0.6131*** | -0.09321 | 0.02571*** | 1,633 | 0.9928 |
| | (0.6574) | (0.365) | (0.164) | (0.1893) | (0.1579) | (0.0003295) | | |

*Notes:* This table reports the results of regressing $\log(HC_{i,t})$ (data) and its model-predicted counterpart on a fourth-order orthogonal polynomial in $\log(\text{AIC}_{i,t})$ plus a year trend, estimated via OLS with standard errors clustered at the country level. The terms $lcop_1$–$lcop_4$ are orthogonal polynomials in $\log(\text{AIC}_{i,t})$ after partialling out the year trend. Significance levels: * p<0.1, ** p<0.05, *** p<0.01.

## 3.4 Results

In Table 3 I present the results of the calibration. The model matches the elasticity from Acemoglu et al. (2013), the trade share from Dingel et al. (2026), and the age expenditure differences in Papanicolas et al. (2020) nearly perfectly. Despite its parsimony, and despite the fact that I only explicitly target Equation 27, the model also does a good job of fitting the entire panel, as shown in Figure 3. In particular, neither the model nor the data indicate that there is any notable curvature in log healthcare expenditures relative to log overall expenditures—both suggest a linear relationship. I formalize this "eyeball test" by regressing both predicted and actual healthcare expenditures on a fourth-order polynomial in AIC. As the coefficients on raw polynomials are unstable, I first orthogonalize the polynomials to avoid multicollinearity. I report the results in Table 4, with the transformed polynomials denoted $lcop1$–$lcop4$ for "log consumption orthogonal polynomials." The model does a good job of matching the data, with the coefficients on the predicted polynomials all contained inside the corresponding confidence intervals on the actual polynomials (there is more variation in the data than the model predicts, hence the wider confidence bands).

As expected, $\rho_h > \rho_n$ and the $\kappa$ shifters are monotone increasing in age, meaning that healthcare and the composite non-healthcare good have own-price elasticities greater than one in absolute value and that healthcare is more productive for the old. I find that healthcare productivity growth is roughly the same as (and perhaps marginally faster than) non-healthcare.

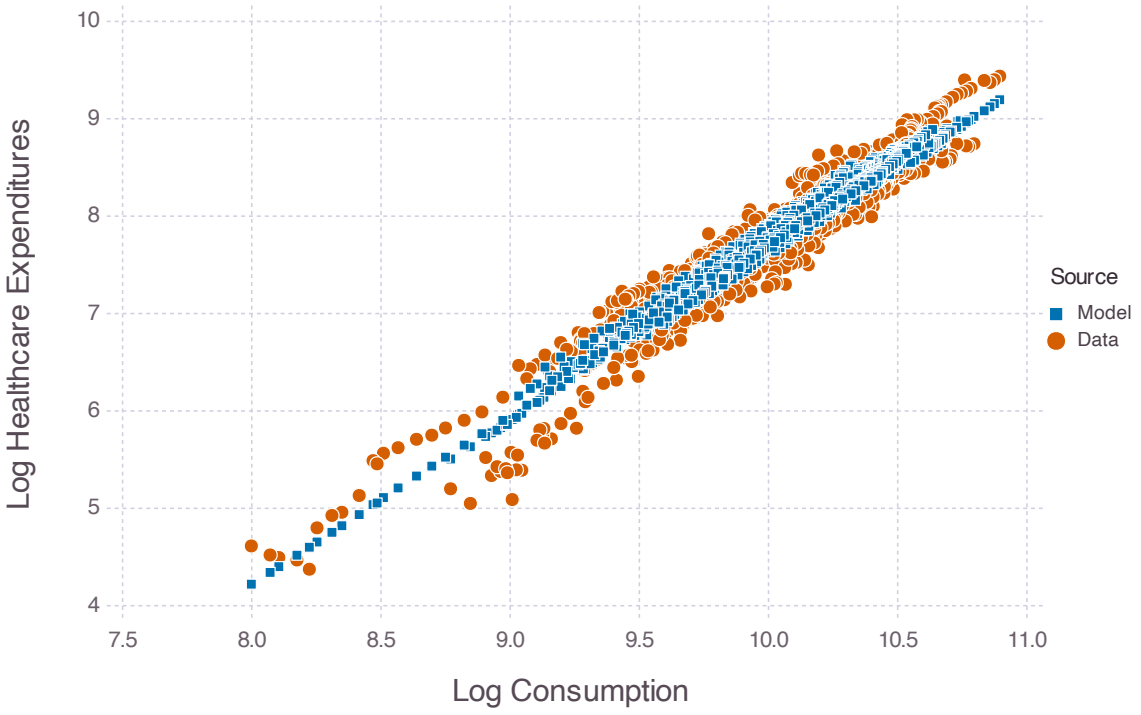



Figure 3: OECD Panel: Model vs Data

*Notes*: This figure compares health spending in the full panel of OECD countries in the real and model-simulated data. Model observations are squares and data observations are circles.

# 4 Counterfactuals

In this section, I use the calibrated model to understand changes in healthcare expenditures across time and place and to explore quantitatively where they may go in the future under the model's maintained assumptions.

## 4.1 Decomposing Past Healthcare Expenditures

The model has three mechanisms with which to explain growth in healthcare expenditures across the OECD: population aging, income growth, and the difference in technological progress between healthcare and non-healthcare. I now use the model to test which explanation is dominant.

First, I keep the age structure of each country's population fixed at its earliest value and simulate a counterfactual panel for healthcare expenditures. Second, I eliminate non-healthcare productivity growth by holding each country's log consumption

fixed at its initial level, while keeping the relative price path between health and non-healthcare unchanged. Third, I set productivity growth in healthcare equal to productivity growth in non-healthcare. I display the results of all these experiments in Figure 4. I find that aging and relative price changes have both increased healthcare expenditures in the OECD panel, but that rising incomes are the dominant explanation. The model infers that the rate of technological progress in healthcare is similar to that in the overall economy, so relative price changes are not a major driver of changes in healthcare expenditures over time. This provides a structural approach to measuring healthcare price changes indirectly, which complements clinical evidence from Dunn et al. (2022) and Dauda et al. (2020). Their measures directly capture health benefits, whereas mine indirectly capture a broad notion of healthcare quality. It is reassuring that we find broadly similar numbers.

## 4.2 Cross-Country Differences in Healthcare Expenditures

I focus on 2024, the most recent year of my data, and study the causes of differences in healthcare expenditures in the cross section. First, I set the age distribution in all countries to the mean in 2024 and then re-solve, so that only differences in productivity drive differences in spending. Next, I set $z_{i,t}$ to its mean level across countries and re-solve for healthcare expenditures, so that only differences in the age distribution drive differences in spending. The results are in Figure 5, with log consumption in the baseline model as the horizontal axis (consumption of course changes in the "Fixed Productivity" case). Differences in incomes are the most important factor. Poor countries tend to be younger, so fixing the age distribution would raise health expenditures in the poorest countries, but the aging-related expenditure differences in Papanicolas et al. (2020) (to which this model is calibrated, cf. Section 3.3) are not large enough to drive large cross-country differences in healthcare expenditures. Hence, the model infers that the productivity shifter $z_{i,t}$ is more important than the age distribution in explaining cross-country differences in healthcare expenditures. This result is robust to recalibrating all six internally estimated parameters to each country profile in Pa-

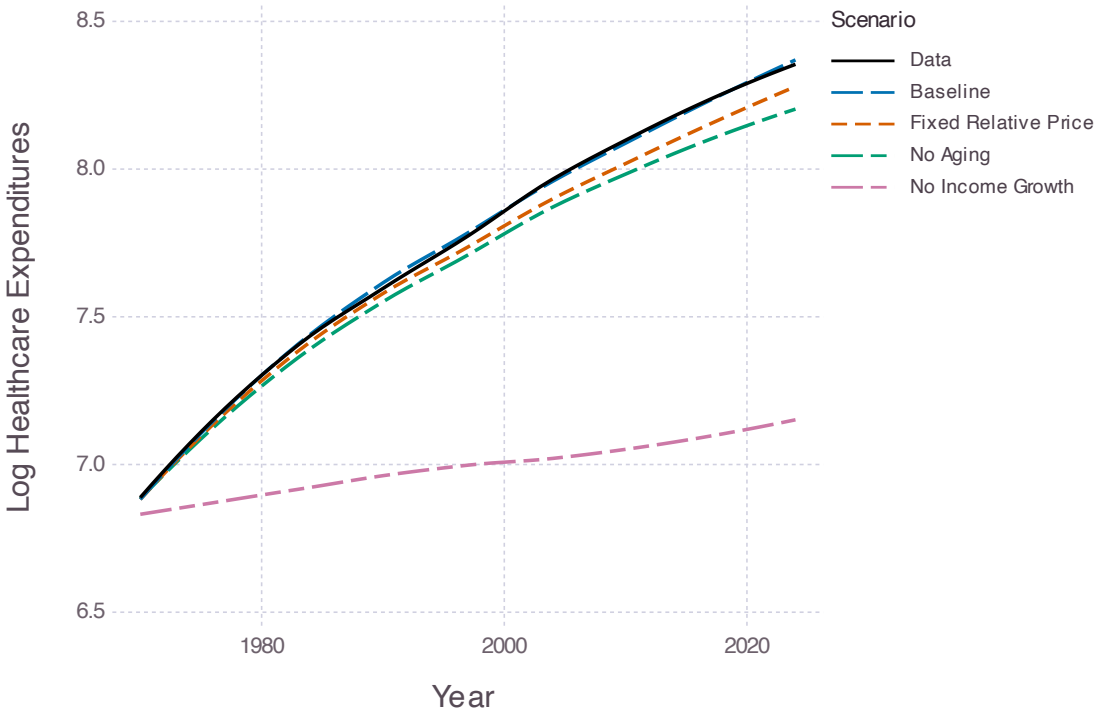



Figure 4: Decomposing Past Healthcare Expenditures

*Notes*: This figure reports mean healthcare spending by year under different counterfactual scenarios in the full OECD panel. "Data" is observed healthcare spending; the "Baseline" scenario comes from the standard model; the "Fixed Relative Price" scenario shows the effect of setting technological growth in healthcare equal to that in non-healthcare and thereby keeping relative prices fixed; the "No Aging" scenario shows the effect of keeping the age distribution in each country at the first level I observe in the data; and the "No Income Growth" scenario shows the effect of setting technological progress in the non-healthcare sector to 0 but keeping $g_h - g_n$ fixed.

panicolas et al. (2020): fixing age reduces the variance of log health spending by 16.9 percent in United States and 18.4 percent in Switzerland, while fixing productivity reduces it by at least 96.9 percent.

## 4.3 Exploring Future Healthcare Expenditures

Finally, through the lens of the model and conditional on its maintained assumptions, I quantitatively explore how aging and productivity growth may influence healthcare expenditures going forward. Analogously to Subsection 4.1, I consider four scenarios: a baseline where population follows the medium projections of United Nations (2024), productivity growth $g_n$ is 2 percent, and $g_h - g_n$ stays at its current value; a second scenario where I freeze the population pyramid in each country; a third scenario where

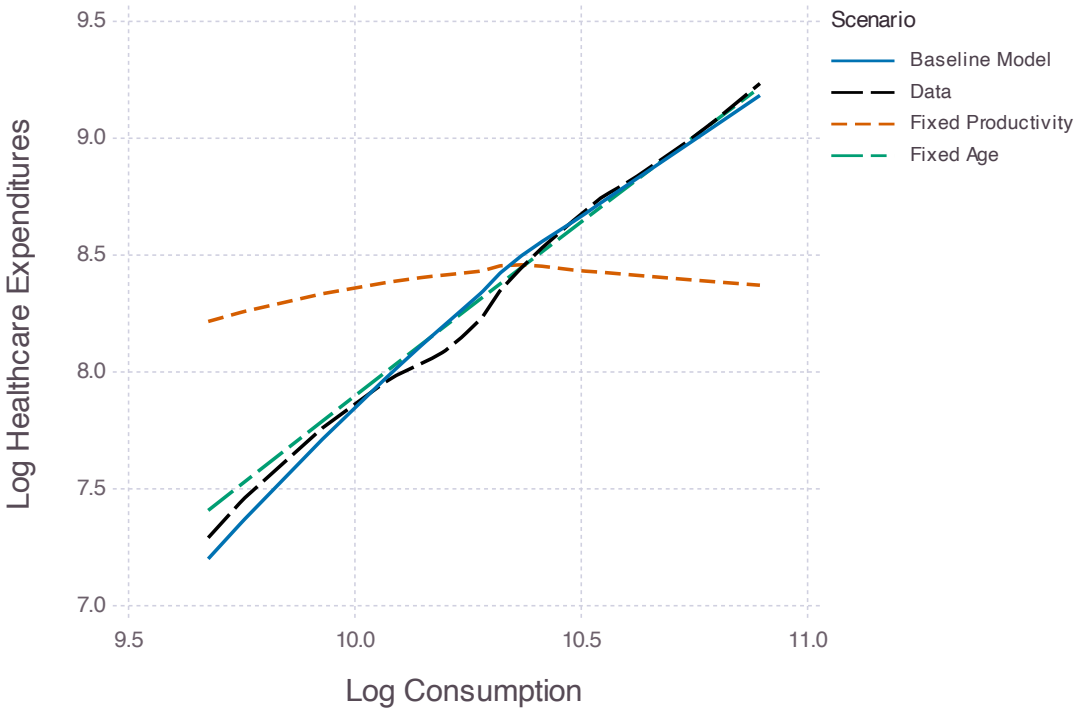



Figure 5: Decomposing Current Healthcare Expenditures

*Notes*: This figure reports healthcare spending under different counterfactual scenarios in the cross section. "Data" is the observed value; the "Baseline" scenario comes from the standard model; the "Fixed Productivity" scenario shows the effect of setting the productivity shifter $z_{i,t}$ equal everywhere; and the "Fixed Age" scenario shows the effect of replacing the age distribution in each country with the average in the data. I use the standard LOESS smoother in Julia's Gadfly package (Jones et al., 2021) to aggregate the data points.

I set $g_n$ to zero but keep $g_h - g_n$ fixed to freeze growth in the non-healthcare sector but keep relative price trends; and a final scenario where I set both $g_h$ and $g_n$ to 2 percent to shut down relative price effects. The results are in Figure 6. Through the lens of the model, both aging and relative price changes increase healthcare expenditures by roughly 5 percentage points of overall consumption by 2101, while rising incomes increase them by around 16 percentage points, making income the dominant force. This mirrors the historical decomposition in Figure 4. The paths look linear through 2101 because this time period only covers an early portion of the transition. For each age group $a$, Equation 7 implies $s_{h,a}^{1-\rho_h}/(1-s_{h,a})^{1-\rho_n} \propto x_a^{\rho_h-\rho_n} p_h^{-\rho_h}$. In the fixed-composition asymptote, $x_a$ grows at $g_n$, so the right-hand side grows at log rate $(\rho_h - \rho_n)g_n + \rho_h(g_h - g_n)$. This is positive in the baseline, so each $s_{h,a}$, and hence the aggregate health share, converges to one. The aggregate share reaches 0.96 in 2500 and

0.9996 in 3000 under a mechanical extrapolation that holds age shares at their 2101 values. This limiting result resembles Jones (2016), where sufficiently strong diminishing marginal utility shifts asymptotically all labor and research toward life-saving technologies, but contrasts with Huetsch et al. (2026), whose endogenous-technology model converges to an interior health share.

One important caveat to the aging-related projections is that they implicitly assume that the relative weights on different age groups in the health planner's calculations will stay the same in the future. Zhao (2014) shows that medical spending has risen faster for the elderly, which he attributes not only to direct increases in public spending on the elderly (e.g. Medicare in the US) but also to ever-more-generous public pension programs (e.g. Social Security in the US) that raise the income of the elderly and therefore their demand for healthcare. If this trend continues, then population aging may have a larger effect on healthcare expenditures than projected in the model.

# 5 Discussion

## 5.1 Productivity Growth Estimates

My model infers that healthcare productivity growth is similar to that in the overall economy, with a 0.3 percentage point per year difference in healthcare's favor at the current calibration. Dunn et al. (2022) find a range of estimates of the change in healthcare costs relative to overall inflation depending on the value of a statistical life year (VSLY). At a VSLY of $50,000, they find that healthcare prices are rising around 0.7 percentage points per year faster than overall prices, with this gap reversing to 1.1 percentage points in healthcare's favor at a VSLY of $100,000 and growing to 3.1 percentage points in healthcare's favor at a VSLY of $150,000. My estimate of healthcare productivity growth is therefore somewhat slower than their central estimate, but well within their range of estimates. Dauda et al. (2020) use a different methodology based on health outcomes in real-world settings and find productivity growth rates for the hospital and nursing home sector of 2.8 percent per year, somewhat greater than my

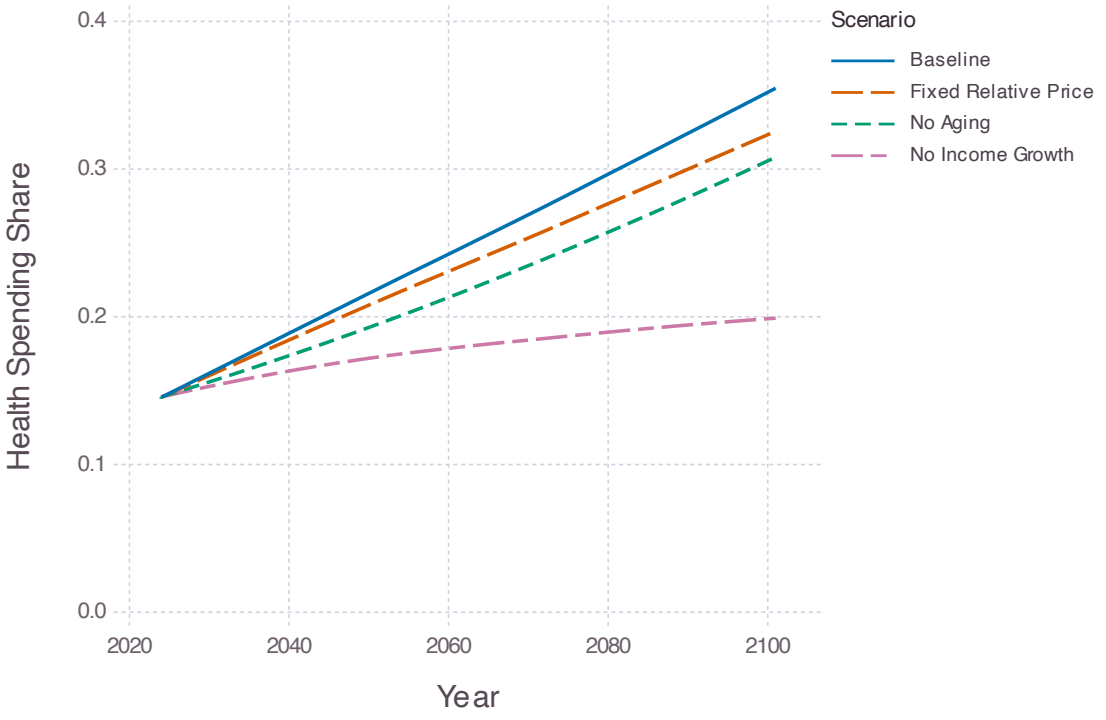



Figure 6: Exploring Future Healthcare Expenditures

*Notes*: This figure reports healthcare spending under different projections in the future. The "Baseline" scenario comes from the standard model; the "Fixed Relative Price" scenario shows the effect of setting technological growth in healthcare equal to that in non-healthcare and thereby keeping relative prices fixed; the "No Aging" scenario shows the effect of freezing the age distribution in each country at its 2024 level; and the "No Income Growth" scenario shows the effect of setting technological progress in the non-healthcare sector to 0 but keeping $g_h - g_n$ fixed. I use the standard LOESS smoother in Julia's Gadfly package (Jones et al., 2021) to aggregate the data points.

estimates, but they acknowledge that this is still sensitive to the choice of VSLY. Reconciling my estimates with those of Dunn et al. (2022) and Dauda et al. (2020) therefore requires a somewhat low but not unreasonable VSLY.

## 5.2 Missing Features

**Imperfect Substitution:** I assume that workers are perfectly substitutable across sectors in both the short run and long run. Imperfect substitutability across sectors would raise the relative price of healthcare in richer countries, because an increase in demand for healthcare would force wages up for the increasingly more reluctant marginal healthcare worker. This can lead to a spurious positive correlation between national income, healthcare prices, and the share of healthcare in expenditures, a cor-

relation that might tempt researchers to conclude that rising prices are the reason why healthcare takes up a large share of expenditures in rich countries. This paper cautions against drawing such conclusions: in this model, it would be more accurate to say that the higher desired healthcare share of expenditures leads to higher relative prices in richer countries. Therefore, income effects need to be even stronger to overcome the drag from rising relative prices. The key fact informing this conclusion is the quasi-experiment of Acemoglu et al. (2013), where a positive productivity shock outside the healthcare sector lowered the healthcare expenditure share. Rising relative prices in healthcare driven by imperfect substitutability of workers should have the same effect.

**Market Power:** The US healthcare market has become more concentrated over time, as documented in Gaynor et al. (2015), which may drive healthcare prices increasingly above marginal costs. If this is the case more generally in the OECD, then technological progress in healthcare needs to be even faster to outpace the increase in relative prices coming from rising markups. That said, I do not find that the US stands out dramatically from the rest of the OECD, except perhaps at the very end of the time series.

**Income Heterogeneity:** I do not account for the fact that incomes vary within regions. If policymakers care about the healthcare demand of the median household, an increase in inequality should push healthcare expenditures down conditional on mean consumption. The model may therefore understate the extent to which the US (a highly unequal country) is an outlier in terms of healthcare expenditures. That said, looking at consumption rather than income is important when considering healthcare expenditures, and consumption is much more equal than income within countries. Hence, this modification seems unlikely to dramatically change the basic results.

**Age-Specific Technological Change:** I assume that technological progress is constant across age groups. However, Zhao (2014) shows that the gradient of healthcare expenditures with respect to age has steepened dramatically since 1960, a fact that my model cannot rationalize without faster technological progress in healthcare for older people compared to younger people. This strengthens the conclusion that aging can-

not explain the rise in healthcare expenditures by itself: even very large shifts in the age distribution would lead to minimal changes in aggregate health expenditures if the gradient of health spending with respect to age remained at its 1960 level.

### 5.3 Policy Implications

The model implies that healthcare is a luxury and that the healthcare spending share rises as healthcare becomes cheaper, in quality-adjusted terms, relative to other goods. This means that bringing down the healthcare spending share requires lowering incomes or *raising* quality-adjusted prices. To give an example of a policy that seemed to succeed in reducing costs, Pinkovskiy (2020) finds that managed-care plans, which restrict the set of doctors that a patient can see, reduced US healthcare spending in the late nineties. He also documents a severe political backlash against these policies, which led to laws limiting their use and driving healthcare spending back up. Pinkovskiy (2020) does not aim to provide a normative analysis of these laws, but through the lens of my model, managed-care plans act as a reduction in quality, or an increase in the quality-adjusted price of healthcare. Pinkovskiy (2020) finds little evidence that such policies affected medical outcomes, but access to a wide network of doctors may make consumers better off for reasons outside of health outcomes[9]. This supports the findings in Ho and Lee (2019) that an insurer's privately-optimal network breadth is smaller than the social optimum. Shepard (2022) offers another mechanism based on adverse selection.

## 6 Conclusion

In this paper, I study why healthcare expenditures are so different across countries and why they have risen as a share of consumption using a model of structural change. I discipline the model with data on healthcare expenditures in a panel of OECD countries, and quasi-experimental evidence on the response of healthcare expenditures to a

[9] Rosenthal (2023) documents significant improvements in hospital amenities, suggesting that productivity growth in healthcare may encompass more than just survival probabilities.

positive productivity shock outside the healthcare sector. The model generates an aggregate income elasticity of healthcare spending above one without contradicting key micro facts that seem in tension with such a high elasticity. I find that rising incomes are indeed the dominant driver of healthcare expenditures across time and place, and that this is likely to continue in the future. In the calibrated model, healthcare demand has an own-price elasticity greater than one in absolute value. As a result, policy changes that improve healthcare-sector productivity and lower its relative price raise healthcare's expenditure share.

### Data Availability Statement

The data used in this paper are publicly available. The data and code needed to reproduce all results are archived on Zenodo: https://doi.org/10.5281/zenodo.22960566.

### Declaration of Generative AI Use

During the preparation of this work, I used OpenAI's ChatGPT and Codex and Anthropic's Claude to review code and prose, correct errors, assist with code development, improve clarity, create visualizations, and check consistency between the code and manuscript. I reviewed and edited all AI-assisted output and take full responsibility for the article.

## References

Daron Acemoglu, Amy Finkelstein, and Matthew J Notowidigdo. Income and health spending: Evidence from oil price shocks. *Review of Economics and Statistics*, 95(4):1079–1095, 2013. doi: 10.1162/REST_a_00306.

Treb Allen and Costas Arkolakis. Trade and the topography of the spatial economy. *The Quarterly Journal of Economics*, 129(3):1085–1140, 2014. doi: 10.1093/qje/qju016.

Gerard F. Anderson, Uwe E. Reinhardt, Peter S. Hussey, and Varduhi Petrosyan. It's the prices, stupid: Why the United States is so different from other countries. *Health Affairs*, 22(3):89–105, 2003. doi: 10.1377/hlthaff.22.3.89.

Sean Bassler. The effect of healthcare costs on low-skill domestic outsourcing. Manuscript, January 2024. URL https://seanbassler.github.io/Bassler_JMP_2023_UMN.pdf.

Timo Boppart. Structural change and the Kaldor facts in a growth model with relative price effects and non-Gorman preferences. *Econometrica*, 82(6):2167–2196, 2014. doi: 10.3982/ECTA11354.

Catherine Boulatoff, Talan B İşcan, and Yulia Kotlyarova. Does distance matter for trade in services? the case of interprovincial trade in Canada. *Open Economies Review*, 33(1):157–185, 2022. doi: 10.1007/s11079-021-09629-3.

Centers for Medicare & Medicaid Services. National health expenditure data: Historical. https://www.cms.gov/data-research/statistics-trends-and-reports/national-health-expenditure-data/historical, 2026a. Accessed 2026-04-29.

Centers for Medicare & Medicaid Services. NHE Fact Sheet, 2026b. URL https://www.cms.gov/data-research/statistics-trends-and-reports/national-health-expenditure-data/nhe-fact-sheet. Page last modified January 14, 2026; accessed May 16, 2026.

Amitabh Chandra and Jonathan S. Skinner. Technology growth and expenditure growth in health care. *Journal of Economic Literature*, 50(3):645–680, 2012. doi: 10.1257/jel.50.3.645.

Diego Comin, Danial Lashkari, and Martí Mestieri. Structural change with long-run income and price effects. *Econometrica*, 89(1):311–374, 2021. doi: 10.3982/ECTA16317.

Teresa A Coughlin, Haley Samuel-Jakubos, and Rachel Garfield. Sources of payment for uncompensated care for the uninsured. Technical report, Kaiser Family Foundation, 2021. URL https://www.kff.org/affordable-care-act/sources-of-payment-for-uncompensated-care-for-the-uninsured/. Accessed September 23, 2026.

David M. Cutler. What is the US health spending problem? *Health Affairs*, 37(3):493–497, 2018. doi: 10.1377/hlthaff.2017.1626.

Seidu Dauda, Abe Dunn, and Anne Hall. Are medical care prices still declining? a systematic examination of quality-adjusted price index alternatives for medical care. *Hutchins Center on Fiscal and Monetary Policy at Brookings Working Paper 59*, 2020. URL https://www.brookings.edu/wp-content/uploads/2020/03/WP59_Dauda-et-al..pdf.

Jonathan I Dingel, Joshua D Gottlieb, Maya Lozinski, and Pauline Mourot. Market size and trade in medical services. Working Paper 31030, National Bureau of Economic Research, July 2026.

Samuel Dodini, Petter Lundborg, Katrine V. Løken, and Alexander Willén. The fatal consequences of brain drain. Research Department Working Paper 2528, Federal Reserve Bank of Dallas, August 2025. URL https://doi.org/10.24149/wp2528.

Stephen W. Duffy, Laszlo Tabar, Anne Helene Olsen, Bedrich Vitak, Prue C. Allgood, Tony H. H. Chen, Amy M. F. Yen, and Robert A. Smith. Absolute numbers of lives saved and overdiagnosis in breast cancer screening, from a randomized trial and from the breast screening programme in england.

*Journal of Medical Screening*, 17(1):25–30, 2010. doi: 10.1258/jms.2009.009094.

Abe Dunn, Anne Hall, and Seidu Dauda. Are medical care prices still declining? a re-examination based on cost-effectiveness studies. *Econometrica*, 90(2):859–886, 2022. doi: 10.3982/ECTA17635.

Fabian Eckert and Michael Peters. Spatial structural change. Working paper, June 2025. URL https://www.fpeckert.me/ssc.pdf.

Keyvan Eslami. *Essays in Health Economics*. PhD thesis, University of Minnesota, 2019.

Eurostat. Glossary: Actual individual consumption (AIC). https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Actual_individual_consumption_(AIC), 2026. Accessed 2026-05-14.

Eurostat, OECD, and World Health Organization. *A System of Health Accounts 2011: Revised Edition*. OECD Publishing, 2017. doi: 10.1787/9789264270985-en.

Tianyu Fan, Michael Peters, and Fabrizio Zilibotti. Growing like India: The unequal effects of service-led growth. *Econometrica*, 91(4):1457–1494, 2023. doi: 10.3982/ECTA20964.

Jesús Fernández-Villaverde, Gustavo Ventura, and Wen Yao. The wealth of working nations. *European Economic Review*, 173:104962, 2025. doi: 10.1016/j.euroecorev.2025.104962.

Raquel Fonseca, Pierre-Carl Michaud, Titus Galama, and Arie Kapteyn. Accounting for the rise of health spending and longevity. *Journal of the European Economic Association*, 19(1):536–579, 2021. doi: 10.1093/jeea/jvaa003.

Raquel Fonseca, François Langot, Pierre-Carl Michaud, and Thepthida Sopraseuth. Understanding cross-country differences in health status and expenditures: health prices matter. *Journal of Political Economy*, 131(8):1949–1993, 2023. doi: 10.1086/724113.

Martin Gaynor, Kate Ho, and Robert J Town. The industrial organization of health-care markets. *Journal of Economic Literature*, 53(2):235–284, 2015. doi: 10.1257/jel.53.2.235.

Thomas E. Getzen. Health care is an individual necessity and a national luxury: Applying multilevel decision models to the analysis of health care expenditures. *Journal of Health Economics*, 19(2):259–270, 2000. doi: 10.1016/S0167-6296(99)00032-6.

Thomas E. Getzen. Aggregation and the measurement of health care costs. *Health Services Research*, 41 (5):1938–1954, 2006. doi: 10.1111/j.1475-6773.2006.00558.x.

Joshua R. Goldstein, Thomas Cassidy, and Kenneth W. Wachter. Vaccinating the oldest against COVID-19 saves both the most lives and most years of life. *Proceedings of the National Academy of Sciences*, 118 (11), 2021. doi: 10.1073/pnas.2026322118.

Robert E. Hall and Charles I. Jones. The value of life and the rise in health spending. *The Quarterly Journal of Economics*, 122(1):39–72, 02 2007. doi: 10.1162/qjec.122.1.39.

John Hejkal, B Ravikumar, and Guillaume Vandenbroucke. Technology adoption, mortality and population dynamics. *The Economic Journal*, 135(666):584–610, 2025. doi: 10.1093/ej/ueae079.

Kate Ho and Robin S Lee. Equilibrium provider networks: Bargaining and exclusion in health care markets. *American Economic Review*, 109(2):473–522, 2019. doi: 10.1257/aer.20171288.

Leon Huetsch, Dirk Krueger, and Alexander Ludwig. The medical expansion, life expectancy, and endogenous directed technical change. Working Paper 35092, National Bureau of Economic Research, April 2026.

Charles I Jones. Why have health expenditures as a share of GDP risen so much? Working Paper 9325, National Bureau of Economic Research, November 2002.

Charles I. Jones. Life and growth. *Journal of Political Economy*, 124(2):539–578, 2016. doi: 10.1086/684750.

Daniel C. Jones, Ben Arthur, Tamas Nagy, Mattriks, Shashi Gowda, Godisemo, Tim Holy, Andreas Noack, Avik Sengupta, Darwin Darakananda, Adam B, Iain Dunning, Simon Leblanc, Rik Huijzer, Keno Fischer, David Chudzicki, Morten Piibeleht, Alex Mellnik, Dave Kleinschmidt, Tom Breloff, Yichao Yu, Joey Huchette, Mike J. Innes, Inkyu, John Verzani, Artem Pelenitsyn, Calder Coalson, Ciarán O'Mara, and Elliot Saba. Giovineitalia/gadfly.jl: v1.3.4, 10 2021. URL https://doi.org/10.5281/zenodo.5559613.

Joseph P. Newhouse. Medical-care expenditure: A cross-national survey. *The Journal of Human Resources*, 12(1):115–125, 1977. doi: 10.2307/145602.

OECD. Annual national accounts: GDP and components—expenditure approach (table 1), 2025a. URL https://data-explorer.oecd.org/.

OECD. OECD health statistics: Health expenditure and financing (database), 2025b. URL https://data-explorer.oecd.org/.

OECD. *Health at a Glance 2025: OECD Indicators*. OECD Publishing, Paris, 2025c. doi: 10.1787/8f9e3f98-en. URL https://doi.org/10.1787/8f9e3f98-en.

Irene Papanicolas, Alberto Marino, Luca Lorenzoni, and Ashish Jha. Comparison of Health Care Spending by Age in 8 High-Income Countries. *JAMA Network Open*, 3(8):e2014688, 08 2020. doi: 10.1001/jamanetworkopen.2020.14688.

Maxim L. Pinkovskiy. The impact of the managed care backlash on health care spending. *The RAND Journal of Economics*, 51(1):59–108, 2020. doi: 10.1111/1756-2171.12306.

Nick Pretnar and Maria Feldman. Health sector structural change. *American Economic Journal: Macroeconomics*, 2026. doi: 10.1257/mac.20240013. Forthcoming.

Random Critical Analysis. Why conventional wisdom on health care is wrong (a primer), 2023. URL https://randomcriticalanalysis.com/why-conventional-wisdom-on-health-care-is-wrong-a-primer/. Accessed September 23, 2026.

Elisabeth Rosenthal. Hospitals have gotten too nice. *The Atlantic*, Nov 2023. URL https://www.theatlantic.com/health/archive/2023/11/hospitals-should-leave-hospitality-hotels/676035/. Accessed September 23, 2026.

Mark Shepard. Hospital network competition and adverse selection: Evidence from the Massachusetts health insurance exchange. *American Economic Review*, 112(2):578–615, 2022. doi: 10.1257/aer.20201453.

United Nations. World population prospects 2024, online edition (dataset): Population 1 January by single age and sex, medium variant, 1950–2100, 2024. URL https://population.un.org/wpp/downloads.

U.S. Office of Management and Budget. Federal receipts as percent of gross domestic product [FYFRGDA188S]. FRED, Federal Reserve Bank of St. Louis, 2023. URL https://fred.stlouisfed.org/series/FYFRGDA188S. Database; accessed September 23, 2026.

Kai Zhao. Social Security and the rise in health spending. *Journal of Monetary Economics*, 64:21–37, 2014. doi: 10.1016/j.jmoneco.2014.02.005.